

Fun with Factors & Multiples

Let's explore numbers together!



What Are Factors?

A factor is a number that divides another number exactly, with no remainder.

Example: Factors of 6 are 1, 2, 3, and 6. This means 6 can be divided equally by each of these numbers without anything left over!



How to Find Factors

01 List the Numbers

Write down all numbers from 1 up to the number you want to find factors for.

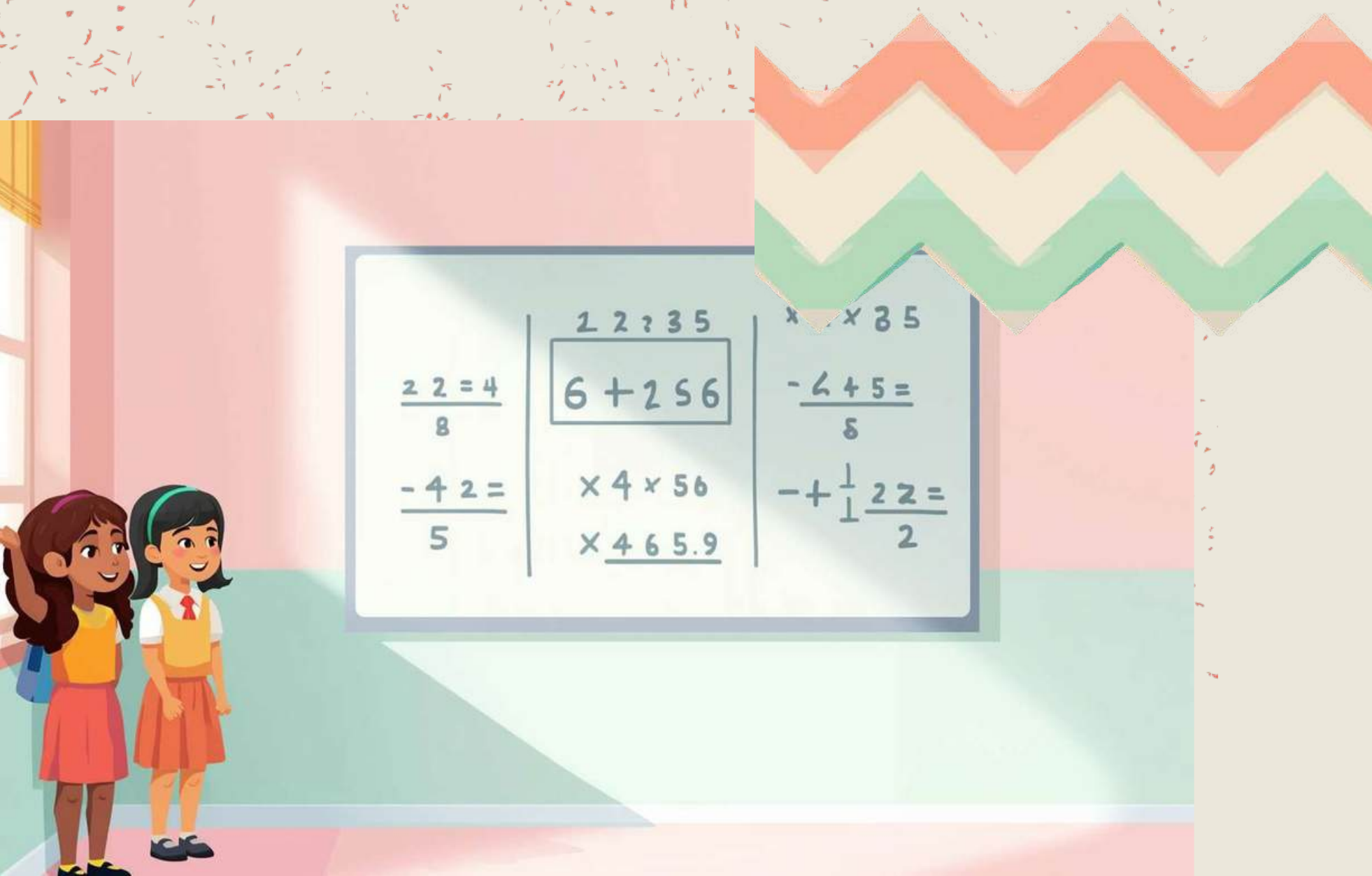
02 Check Division

Test each number to see if it divides exactly with no remainder left over.

03 Example: Factors of 8

$8 \div 1 = 8 \checkmark$, $8 \div 2 = 4 \checkmark$, $8 \div 4 = 2 \checkmark$, $8 \div 8 = 1 \checkmark$

So factors of 8 are: 1, 2, 4, 8



Practice Time!

Find the Factors

YOUR CHALLENGE

What are all the factors of 10? Think about which numbers divide 10 exactly!

CHOOSE THE RIGHT ANSWER

A) 1, 2, 5, 10 B) 1, 3, 5 C) 2, 4, 10

THINK & SELECT!

Remember: Factors divide a number exactly with no remainder. Try dividing 10 by each option!



What Are Multiples?

A multiple is a number you get when you multiply a number by 1, 2, 3, and so on.

Example: Multiples of 3 are 3, 6, 9, 12, 15...
Think of it like skip counting! When you count by 3s, you are listing the multiples of 3.
Multiples go on forever because you can keep multiplying!



0 — 2 — 4 — 6 — 9 — 11 — 12 — 17 — 13

How to Find Multiples



01 Multiply Step by Step

To find multiples, multiply the number by 1, then by 2, then by 3, and keep going!

02 Example: Multiples of 4

$4 \times 1 = 4$, $4 \times 2 = 8$, $4 \times 3 = 12$, $4 \times 4 = 16$... So multiples of 4 are 4, 8, 12, 16, 20...

03 Continue the Pattern

Multiples go on forever! Just keep multiplying by the next number. Try finding multiples of 5: 5, 10, 15, 20, 25...



Practice Time!

Find the Multiples

THE QUESTION

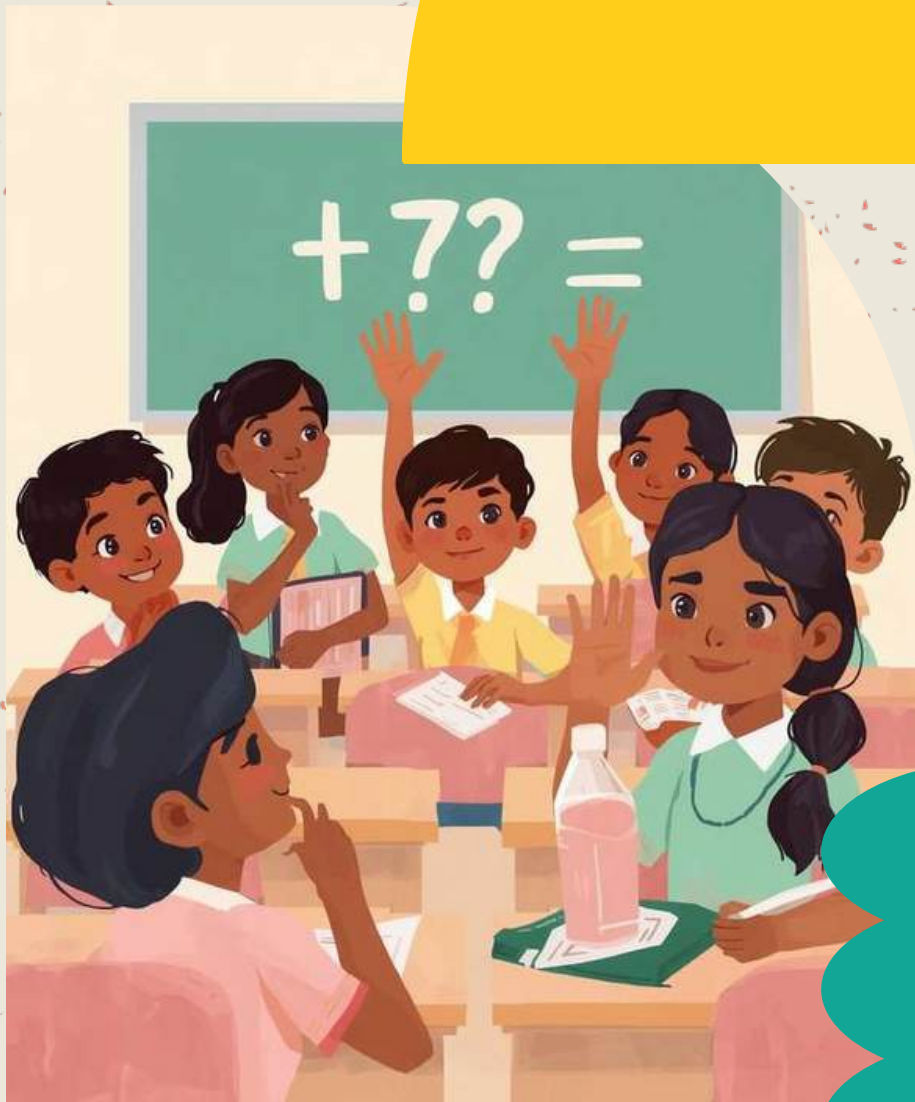
Which of these numbers are multiples of 5?
Think carefully before you answer!

YOUR OPTIONS

5, 10, 12, 15 — Circle all the multiples of 5.
Remember: 12 is NOT a multiple of 5!

CORRECT ANSWERS

The multiples of 5 are: 5 ✓, 10 ✓, and 15 ✓. Great job if you got them all right!



FACTORIAL

How 6 crrr have recn crvied l carning l can in divieot egantly.

$$\begin{array}{c} \text{Apple} \\ \times \\ \text{Apple} \end{array} + \begin{array}{c} \text{Apple} \\ \times \\ \text{Apple} \end{array} = \begin{array}{c} \text{Apple} \\ \times \\ \text{Apple} \end{array}$$

$$\begin{array}{c} \text{Apple} \\ \times \\ \text{Apple} \end{array} + \begin{array}{c} 6 \\ \times \\ 5 \end{array} = \begin{array}{c} \text{Apple} \\ \times \\ \text{Apple} \end{array}$$

$$\begin{array}{c} \text{Apple} \\ \times \\ \text{Apple} \end{array} + \begin{array}{c} \text{Apple} \\ \times \\ 2 \end{array} = \begin{array}{c} \text{Apple} \\ \times \\ \text{Apple} \end{array}$$

FACTORS VS MULTIPLES: SPOT THE DIFFERENCE!

Factors are numbers that divide another number exactly with no remainder. Example: Factors of 6 are 1, 2, 3, 6

MULTIPLES ARE DIFFERENT: THEY KEEP GROWING!

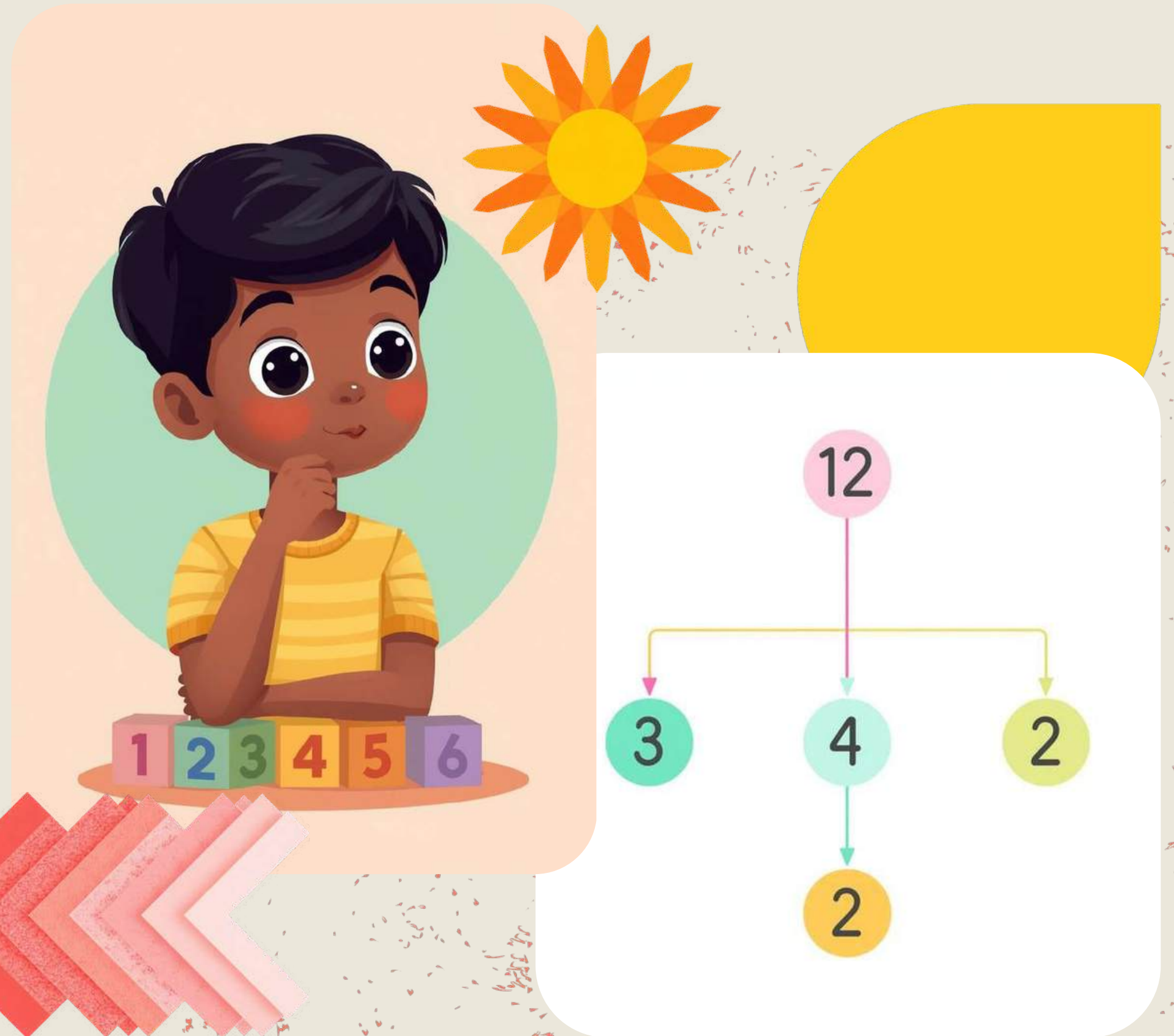
Multiples are numbers you get when you multiply a number by 1, 2, 3, and so on.
Example: Multiples of 6 are 6, 12, 18, 24...

$$\begin{array}{l} 6 \rightarrow 12 \rightarrow 6 \\ 18 \rightarrow 18 \rightarrow 24 \\ 6 \longrightarrow 24 \end{array}$$

Factor Trees

A fun way to break numbers into smaller factors step by step!

Factor trees help us find all the factors of a number by splitting it into smaller parts. Start with a number, find two factors that multiply to make it, then keep breaking down until you reach prime numbers. It's like a family tree for numbers!





PRIME AND COMPOSITE NUMBERS

Prime Numbers have only two factors: 1 and the number itself.
Examples: 2, 3, 5, 7, 11

COMPOSITE NUMBERS: MORE THAN TWO FACTORS

Composite Numbers have more than two factors. They can be divided by other numbers too!

Examples: 4, 6, 8, 9, 12



Why Are Factors & Multiples Important?

SHARING EQUALLY

Factors help us share things fairly! If you have 12 candies for 4 friends, factors tell you each gets 3.

COUNTING IN GROUPS

Multiples help us count in groups! Skip counting by 5s uses multiples: 5, 10, 15, 20...

REAL-LIFE MATH

From arranging desks in rows to dividing snacks at a party, factors and multiples are everywhere in daily life!



Summary: What We Learned

Let's review the key points from today's lesson:

Factors divide numbers exactly with no remainder

Multiples are results of multiplication

Factor trees help us find all factors step by step

Prime numbers have only 2 factors, composite have more



Challenge Quiz: Let's Test Your Skills!

01 Is 7 a factor of 28?

Think carefully! Does $28 \div 7$ give a whole number with no remainder? Circle your answer: Yes or No

02 Is 15 a multiple of 5?

Remember, multiples are what you get when you multiply! Can you get 15 by multiplying 5? Yes or No

03 Find one factor of 18

Which numbers divide 18 exactly? Write down one factor you can find. Hint: Start checking from 1, 2, 3...



Keep Exploring!

Great job learning about factors and multiples! You're becoming a maths superstar!

Here are fun ways to practice more:

- Count objects around you and find their factors
- Play skip counting games with friends
- Draw factor trees for different numbers
- Challenge yourself with bigger numbers!



Thank You for Learning With Us!

HAVE QUESTIONS? KEEP EXPLORING FACTORS AND MULTIPLES!



Extra Practice (Optional)

FIND FACTORS OF 15

Which numbers divide 15 exactly? Hint: Start with 1 and work your way up!

MULTIPLES OF 7

List the first 5 multiples of 7. Remember: multiply 7 by 1, 2, 3, 4, and 5!

PRIME OR COMPOSITE?

Is 11 a prime number or a composite number? Think about how many factors it has!

